Pratham Aggarwal

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Education

2024 – 2027 **University of California San Diego**, La Jolla, CA

B.S. in Data Science

Honors: HDSI Scholar; HKN-IEEE.

Research Interests: Machine Learning, Climate Science, Probability & Statistics.

Experience

Research

2025–Present **Undergraduate Researcher**, *Climate Analytics Lab*, San Diego, CA

Advisor: Duncan Watson-Parris

- Developed FASST, a coupled climate model that replaced the computationally expensive atmosphere and sea ice with lightweight U-Net emulators while retaining the prognostic POP2 ocean. Demonstrated stable climate behavior in a 140-year unforced simulation with realistic interannual ENSO variability.
- Built a 7.7M-parameter FASST emulator achieving 160.9 simulated years/day with R^2 above 0.9 on most fields.
- Developed a diffusion model for AIMIP to predict 6-hourly atmospheric fields from sea-surface temperature and solar.
- Contributed to ClimateBench 2.0, a benchmark of 14 CMIP6 climate models using MAE, RMSE, and CRPS to evaluate historical and future climate predictions; received a $4,500 HDSI scholarship for the project.

2026–Present **Undergraduate Researcher**, *Belkin Lab*, San Diego, CA

Advisor: Qingsong Wang

- Researching activation steering for controllable refusal in masked diffusion language models (LLaDA-8B); extracted topic-specific and global refusal directions via single-pass CAA across all 32 layers.
- Used layer-wise probing to show topic identity is linearly decodable (54% vs. 25% chance); found topic directions are geometrically distinct, with self-harm near-orthogonal to global refusal.

2025 **Undergraduate Researcher**, *Existential Robotics Lab*, San Diego, CA

Advisor: Nikolay A. Atanasov

- Evaluated A*, RRT, and MPC motion-planning algorithms in dynamic environments, reducing path-planning failures by 28% while maintaining average trajectory error below 0.15 m.
- Increased simulation speed by 2.1× by parallelizing physics evaluations and testing motion-planning algorithms concurrently.

Industry

2026 **AI/ML Research Intern**, *NIOV Labs*, San Diego, CA
- Reduced API costs by 23% by building a task-routing system that assigned requests to AI agents based on task complexity using prompt-based classification.
- Improved evaluation of AI responses by designing an LLM-as-a-judge system with a detailed rubric to score responses on tone, length, and content against target requirements.

2025 **Data Science Intern**, *Solana Center*, San Diego, CA
- Led a 5-member team to analyze composting records using Python and SQL and built a real-time Power BI dashboard to track key metrics and support business decisions, contributing to a 5% increase in business growth.
- Built automated Python routines for data cleaning, manipulation, and missing-value imputation, making data preparation faster, consistent, and reproducible across analyses.

2025 **Quantitative Developer Intern**, *SFIC Quantitative Technologies*, San Diego, CA
- Increased simulated options trading returns by 12% for a $1.3M student-run investment fund by building a deep learning model to select option strategies and exercise timing.
- Improved simulated trading performance by 18% by building a model that selected option type, expiration, and exercise timing from historical market data.

Publications

NeurIPS 2026 **P. Aggarwal**, D. Watson-Parris, and W. E. Chapman. FASST: A Compact Stateless Atmosphere Model for Long-Term Ocean Coupling. In *Tackling Climate Change with Machine Learning Workshop at NeurIPS*, 2026. arXiv:2610.36297.

Presentations

HDSI 2025 **P. Aggarwal**, K. Shah, W. Tobin, and D. Watson-Parris. ClimateBench 2.0: Probabilistic Climate Model Scoring. Poster presented at the Halıcıoğlu Data Science Institute (HDSI), UC San Diego, 2025. Web app: climate-analytics-lab.github.io/ClimateBench_app.

Projects

2026 **Coastal Flooding Prediction Under Out-of-Distribution Conditions**, *NSF Hackathon, 1st Place*
- Improved coastal flooding prediction F1 score from 0.53 to 0.94 by testing XGBoost, LightGBM, Random Forest, LSTM, and Transformer models and selecting CatBoost for the final model.
- Improved model performance through iterative feature engineering, testing features informed by coastal flooding physics during a 48-hour hackathon.

2025 **Homeless Services Computer Vision Platform**, *Brain Corp Hackfrontier, 1st Place*
- Improved homeless service center placement by building a geospatial model using 35+ transit, demographic, and geographic features, achieving 67% placement accuracy.
- Built a real-time computer vision system using Oxen.ai and EyePop.ai to track changes in homeless population locations and support service center placement decisions.

Skills

Languages	Python, SQL, C, C++, Java
AI / ML	PyTorch, TensorFlow, Scikit-Learn, Pandas, Matplotlib, Power BI, Tableau
Systems & Tools	Docker, Kubernetes, AWS, Google Cloud, Git/GitHub, HPC (NCAR Derecho, Casper; NSF NRP)